

Birthday attack-modeling

Module IV



The Birthday attack problem modeling

- Assume there are 30 students in a classroom. Let us try to answer the likeliness of the occurrence of the events A and B.

event A : Two people among the 30 students share their birthdays.

event B : No two people in the room has the same birthday.

Modeling the problem:

Assume there are 365 days (leap year ignored). Each person is being randomly assigned a birthday (i.e. any day is equally likely assigned to the birthday of any student).

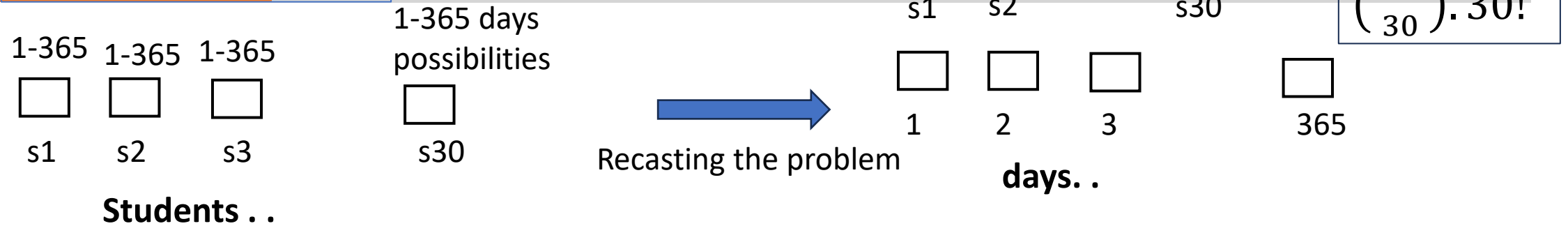
How do we compute the probability that any two students share their birthdays?

- 2 students may share their birthdays in many ways.
- 15 students may share their birthdays in many ways.
- 30 students may share their birthdays in many ways.

and so on. The complexity of computing is high. Hence, computing complement event, rather easy.

The Birthday attack problem modeling

Modeling the problem:



For every student, the probability of assigning each birth date out of 365 days is equally likely.

For the 365 days, the probability of assigning any 30 birth Dates (say $d1, d2, \dots, d30$) to 30 students in different ways.

Number of ways picking birthdays $d1, d2, \dots, d30$ out of 365 days is $\binom{365}{30}$

Then assigning these chosen 30 birthdates to any one permutation of $30!$ is $\binom{365}{30} \cdot 30!$

Total sample space is $365 \cdot 365 \cdot 365 \dots 365 = 365^{30}$

Total number ways of arranging people, so that no two of them share their birthday is $\frac{\binom{365}{30} \cdot 30!}{365^{30}}$.

The Birthday attack cntd..

[illegible]

$$= \frac{\binom{365}{30} \cdot 30!}{365^{30}} = 1 \cdot \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \left(1 - \frac{3}{365}\right) \cdots \left(1 - \frac{29}{365}\right).$$

First student, one date is assigned, probability is 100%, 1

Second student date assignment probability, $1 - \frac{1}{365}$

Third student date assignment probability, $1 - \frac{2}{365}$

and so on ..

The Birthday attack cntd..

$$\boxed{= \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \left(1 - \frac{3}{365}\right) \cdots \left(1 - \frac{29}{365}\right)} \quad \longrightarrow \quad \prod_{i=1}^{p-1} \left(1 - \frac{i}{365}\right)$$


The equation further can be reduced by considering the Taylor Series:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \cdots$$

If $x \ll 1$, then $e^{-x} \approx 1 - x$.

Therefore, the probability that no collision occurs if there are p random elements may be rewritten as:

$$p(\text{no collision}) \approx \prod_{i=1}^{p-1} e^{-\frac{i}{n}} \approx e^{-\frac{1}{n}} \cdot e^{-\frac{2}{n}} \cdot e^{-\frac{3}{n}} \cdots e^{-\frac{(p-1)}{n}} \approx e^{-\frac{(1+2+3+\dots+p-1)}{n}} \approx e^{-\frac{p \cdot (p-1)}{2n}}$$

The Birthday attack cntd..

- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) \approx e^{-\frac{p \cdot (p-1)}{2n}}$$

Now, the probability of that at least one collision occurs determined as,

$$p(\text{at least one collision}) \approx 1 - p(\text{no collision})$$

$$p(\text{at least one collision}) \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

If ε (*epsilon*), is used to denote the probability that at least one collision occurs, then:

$$\varepsilon \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

The Birthday attack cntd..

$$\rightarrow \varepsilon \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

$$\rightarrow e^{-\frac{p \cdot (p-1)}{2n}} \approx 1 - \varepsilon$$

$$\rightarrow -\frac{p \cdot (p-1)}{2n} \approx \ln(1 - \varepsilon)$$

$$\rightarrow p \cdot (p - 1) \approx -2n \cdot \ln(1 - \varepsilon)$$

$$\rightarrow p^2 - p \approx 2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)$$

For large values of p , i.e. a large number of random elements to be tested to find a collision, p^2 is significantly large than p .

Therefore,

$$p^2 \approx 2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)$$

Therefore, the probability that at least one collision occurs is ,

$$p \approx \sqrt{2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)}$$